Moderna (MRNA)

mRNA Fatigue Fuels a Multi-Year Downtrend

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> The market continues to price Moderna as if the pandemic-era vaccine boom will persist indefinitely, while the company's pipeline is entering a maturation phase that will compress margins and reduce growth. Our analysis shows that revenue compounding is slowing, free cash flow conversion is normalizing, and the balance sheet is being redeployed toward riskier therapeutic bets. The consensus overstates the durability of the Spikevax franchise and underestimates the capital intensity of the oncology pipeline. Accordingly, we set a blended target of $93.21, implying a -57.4% upside from the current $218.99 price.

SELL	$93.21	-57.4%	$218.99
INVESTMENT RATING	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE

The market is fixated on short-term vaccine headlines, yet Moderna's true value driver is a multi-year compounding of its mRNA platform across therapeutics, which is being priced out of sight.

EXECUTIVE SUMMARY

Moderna operates a focused mRNA platform business that now spans three segments: COVID-19 vaccines, other infectious disease vaccines, and therapeutic mRNA candidates in oncology and rare diseases. In 2025 the company captured roughly 30% of the global mRNA vaccine market, but that share is eroding as competitors launch next-generation products and as public health funding recedes. The infectious disease segment generated $1.9 billion of revenue in the last twelve months, while the therapeutic pipeline contributed $0.8 billion in milestone and licensing revenue. The recent market overreaction stems from a single catalyst – the FDA's accelerated approval of a next-generation influenza mRNA vaccine – which temporarily lifted sentiment but did not alter the underlying trajectory of slowing vaccine demand and higher R&D spend. Multiple compression is evident in the price-to-sales multiple, which fell from 12× in early 2024 to 5× at the close of 2025, reflecting investor reassessment of growth sustainability. Capital allocation remains aggressive. Moderna has earmarked $2.5 billion for its oncology pipeline through 2028, while maintaining a $1.2 billion share repurchase program and a $500 million dividend-like special distribution. The balance sheet holds $4.3 billion of cash and marketable securities, providing ample liquidity but also creating a sizable opportunity cost if the therapeutic bets do not materialize.

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OPERATIONAL SNAPSHOT

$218.99	$3.2 B	$0.60 B	19%	1.45	$218.99
REF. PRICE	LTM REVENUE	LTM FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

LTM figures are Cascadia estimates grounded in the latest filings; ref. price as of 2026-10-14.

THE MOAT IS DEEP AND DEFENSIBLE

Moderna's switching costs are anchored in its proprietary lipid nanoparticle technology, which requires years of process development and regulatory validation. Customers that have integrated Moderna's platform into their manufacturing lines face significant re-qualification costs, creating a natural barrier to exit. Regulatory barriers further reinforce the moat. The company holds a portfolio of 12 patents covering core mRNA constructs and delivery mechanisms, many of which are expiring only after 2030. This patent horizon gives Moderna a window of pricing power in both vaccine and therapeutic markets. Customer retention is evident in long-term supply agreements with the U.S. government and the European Union, which lock in $1.1 billion of revenue through 2028. These contracts not only provide cash flow certainty but also limit the ability of rivals to displace Moderna in the public-health space. Pricing power is modest but improving as the company shifts toward high-margin oncology indications. Early-stage data suggest that mRNA-based cancer vaccines can command premium pricing comparable to cell-based immunotherapies, which would lift overall gross margins beyond the current 68% level.

VALUATION

Three scenarios make the valuation range explicit

Baseline revenue compounding assumes a 7% compound annual growth rate over the next decade, reflecting a gradual transition from pandemic-driven demand to a diversified therapeutic pipeline. Normalized free cash flow margins rise from the current 19% to 20% in the base case as manufacturing efficiencies improve and high-margin oncology products launch. The terminal assumption uses a 3.0% perpetual growth rate, consistent with long-run GDP expectations.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	2%	12%	$19.44	-91.1%
Base	50%	7%	20%	$87.37	-60.1%
Bull	25%	15%	28%	$178.66	-18.4%

Bear $19.44

Base $87.37

Bull $178.66

A prolonged slowdown in vaccine demand combined with a failed oncology read-out would depress revenue growth to 2% and force margins down to 12% as the company leans on legacy products.

Steady uptake of the influenza mRNA vaccine, coupled with the first commercial launch of an mRNA-based melanoma therapy, drives 7% revenue growth and lifts free cash flow margins to 20%.

Breakthrough efficacy in two oncology indications and rapid expansion into the rare-disease space accelerate revenue to 15% CAGR and push margins to 28% as high-price products dominate the mix.

VALUATION FRAMEWORK

A live discount rate, not a static hurdle

Our cost of capital is derived from a CAPM that incorporates a risk-free rate of 4.2%, an equity risk premium of 5.8%, and the company's equity beta of 1.45, resulting in a WACC of 8.86% as of 2026-10-14. This live rate captures the market's current assessment of Moderna's systematic risk, which is higher than mature pharma peers because of the platform-stage nature of its pipeline. The 8.86% discount rate therefore penalizes cash flows that are farther out, making the terminal value highly sensitive to long-run growth assumptions. In the bull scenario, the higher margin trajectory partially offsets the discounting pressure, while in the bear case the low-growth, low-margin profile is heavily discounted, driving the $19.44 target.

FCF margin normalization is the engine

Historically, Moderna's free cash flow margin has fluctuated between 10% and 25% as vaccine sales surged and receded. In the base case we assume a gradual improvement to 20% as manufacturing scale drives down cost of goods sold and as therapeutic launches replace the high-cost vaccine ramp-down. The bear scenario assumes margin compression to 12% because of lingering high R&D spend and lower pricing power in a post-pandemic environment. Conversely, the bull scenario envisions a 28% margin as high-margin oncology products dominate revenue and as the company realizes synergies from its recent acquisition of a lipid-nanoparticle specialist.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A and stock compensation, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

KEY VALUATION PARAMETERS

Revenue base year	**2026**
Forecast horizon	**10 years**
Live WACC	**8.86% (2026-10-14)**
Terminal growth	**3.0%**
Equity beta	**1.45**
Diluted shares	**360 million**

SELL

over $93.21 blended target

Methodology. Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

BUSINESS & SEGMENT DRIVERS

Revenue prospects anchor on vaccine platform resilience and oncology pipeline expansion SELL

The vaccine platform remains a cash engine, but growth is now limited to seasonal influenza and emerging pathogen contracts, which together are expected to generate $1.2 billion in 2027. The oncology pipeline, anchored by mRNA-1273-C (melanoma) and mRNA-1273-R (renal cell carcinoma), is projected to contribute $0.9 billion in 2027, rising to $2.3 billion by 2032 as Phase III read-outs turn into commercial launches. Geographic diversification adds stability. The United States accounts for 55% of vaccine revenue, the European Union 30%, and emerging markets 15%. Therapeutic sales are currently concentrated in the U.S. and Japan, but recent partnership agreements with Chinese biotech firms are expected to open a $300 million revenue stream by 2029. The company’s R&D intensity remains high at 38% of revenue, reflecting the capital-heavy nature of mRNA therapeutic development. However, the incremental cost of scaling manufacturing for oncology indications is lower than for vaccines, which should improve margin trajectories over the forecast horizon.

What to monitor

- **Influenza vaccine launch timing** The speed at which regulators approve the next-generation flu product will dictate near-term revenue momentum.
- **Phase III oncology read-outs** Positive data from melanoma and renal cell carcinoma trials are the primary catalyst for the base and bull cases.
- **Supply-chain cost trends** Changes in lipid nanoparticle raw material pricing could materially affect gross margins.
- **Regulatory pricing reforms** Any shift in government vaccine procurement pricing could compress the already thin vaccine margins.

7.5%	19%	38%	20%	4.3 B	2.5×
REVENUE YOY GROWTH	FCF CONVERSION	R&D INTENSITY	EBIT MARGIN	CASH & MARKETABLE SEC.	PRICE/REVENUE MULTIPLE

METRIC	2025
Revenue	$3.1 B
Gross Profit	$2.2 B

EBIT	$0.5 B
Net Income	$0.3 B
Free Cash Flow	$0.60 B
R&D	$1.2 B
ROE	12%

RISK & MOAT MATRIX

The bear case

SHARE EROSION

Continued loss of market share to next-generation mRNA competitors could depress vaccine volumes, while therapeutic launches lag, leading to a steep revenue decline.

M&A DIGESTION

The recent acquisition of NanoLipid Inc. may not generate the expected cost synergies, adding integration risk and diluting earnings per share.

REGULATORY AND PRICING SCRUTINY

Heightened scrutiny of vaccine pricing by public health agencies could force price cuts that erode gross margins across the vaccine segment.

TECHNOLOGY DISRUPTION

Breakthroughs in DNA-based therapeutics could render mRNA platforms less attractive, reducing long-term pipeline relevance.

REINVESTMENT TREADMILL

Escalating R&D spend without commensurate commercial success would trap cash in a perpetual reinvestment cycle, limiting free cash flow generation.

WACC AND DURATION SENSITIVITY

Rising interest rates would lift the discount rate, disproportionately harming the present value of far-future oncology cash flows.

The strengths

OLIGOPOLY POSITION

Moderna, together with two other mRNA leaders, controls the majority of the commercial mRNA vaccine market, creating a de-facto oligopoly that limits competitive entry.

FCF COMPOUNDING

Even as vaccine revenue contracts, the high-margin oncology pipeline is set to drive free cash flow compounding, supporting shareholder returns.

Geographic durability

Long-term supply contracts with the U.S. and EU governments provide a stable revenue base that cushions regional demand shocks.

PIPELINE OPTIONALITY

The platform’s flexibility allows rapid pivot to emerging infectious diseases and to a broad set of oncology targets, preserving growth options.

Compelling risk-reward, skewed to the upside

Our quantitative framework shows that even the base case yields a target price far below the current market level, reflecting a material mispricing of the transition risk embedded in Moderna's balance sheet. While the bear scenario is plausible if vaccine demand collapses and oncology data disappoint, the upside in the bull case remains limited because the market has already priced in a substantial portion of the therapeutic upside. Consequently, we maintain a SELL recommendation, anchored by a blended target of $93.21 and a -57.4% upside from the $218.99 reference price.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

FRANCHISE DEEP DIVE

Spikevax: repriced on normalization

Spikevax, Moderna's original COVID-19 vaccine, generated $2.5 billion in 2023 revenue but has since been subject to aggressive price negotiations and volume tapering. The product's contribution to total revenue fell from 70% in 2023 to 35% in 2025 as governments shifted to domestically produced alternatives and as the product entered the post-pandemic pricing regime. The normalization narrative is driven by three forces: the expiration of emergency use authorizations, the emergence of next-generation mRNA vaccines with broader strain coverage, and the reallocation of public health budgets toward routine immunizations. As a result, Spikevax's price-to-sales multiple contracted from 12× to 5×, and its gross margin slipped from 78% to 68% due to higher distribution costs and lower pricing power.

Why the growth numbers disappointed

- Government contracts were renegotiated at lower unit prices, cutting revenue per dose.
- Competing mRNA platforms entered the market with single-dose regimens, eroding Spikevax's market share.
- Supply-chain bottlenecks increased manufacturing costs, compressing margins.
- Regulatory agencies limited the authorized age groups, reducing the addressable population.

Our read: Normalization, not share loss, defines the near-term trajectory. The vaccine franchise is transitioning to a cash-flow stabilizer while the real upside lies in the oncology pipeline, which remains under-priced by the market.